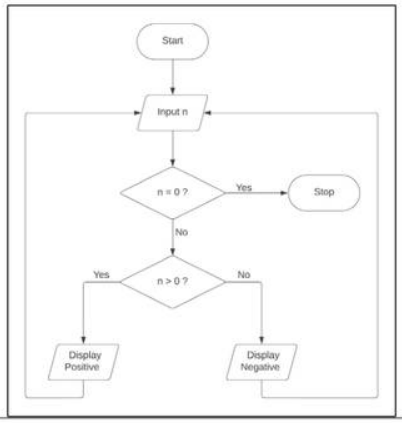
Type MS · Write an algorithm on paper: flowcharts and pseudocode

Answer 1 of 6 · Question 6 part a · 6 marks

2025 JUN · Paper 01 · extract 1 of 4 · find part a in this mark scheme extract

Question	Answer	Additional Guidance	mark
6(a)	Award **one** mark for each of: • Start and Input n identified (accept number or num) (1) • Decision box test for n = 0/n<>0 with yes & no/true & false labels that correctly lead to End/Stop (1) • Decision box test for n > 0 or n<0 with yes & no/true & false labels (1) • Correct messages output including leads from correct yes & no/true & false (1) • Flowchart is fully connected and functionally correct (1) • Correct flowchart symbols used throughout for terminators, input/output and decision boxes (1) Example:  	Accept == for MP2 For MP2 and MP3 decision boxes must have yes/no or true/false labels. For MP5 there must be arrows on connections	6

Type MS · Write an algorithm on paper: flowcharts and pseudocode

Answer 1 of 6 · Question 6 part a · 6 marks

2025 JUN · Paper 01 · extract 2 of 4 · find part a in this mark scheme extract

<table>
<tr><th>Question</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>6(b)(i)</td><td>Award one mark for each pair of columns:
<table><tr><th>number</th><th>flag</th><th>testNum</th><th>DISPLAY</th></tr><tr><td>7</td><td>FALSE</td><td>1</td><td></td></tr><tr><td></td><td></td><td>2</td><td></td></tr><tr><td></td><td></td><td>3</td><td></td></tr><tr><td></td><td></td><td>4</td><td></td></tr><tr><td></td><td></td><td>5</td><td></td></tr><tr><td></td><td></td><td>6</td><td></td></tr><tr><td></td><td>TRUE</td><td>7</td><td></td></tr></table></td><td>Accept for first pair
<table><tr><th>number</th><th>flag</th></tr><tr><td>7</td><td>FALSE</td></tr><tr><td>7</td><td>FALSE</td></tr><tr><td>7</td><td>FALSE</td></tr><tr><td>7</td><td>FALSE</td></tr><tr><td>7</td><td>FALSE</td></tr><tr><td>7</td><td>FALSE</td></tr><tr><td>7</td><td>TRUE</td></tr></table>Do not award either mark point if first row is 7 FALSE 2</td><td>2</td></tr>
<tr><th>Question</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>6(b)(ii)</td><td>Award one mark for each pair of columns:
<table><tr><th>number</th><th>flag</th><th>testNum</th><th>DISPLAY</th></tr><tr><td>9</td><td>FALSE</td><td>1</td><td></td></tr><tr><td></td><td></td><td>2</td><td></td></tr><tr><td></td><td>TRUE</td><td>3</td><td></td></tr><tr><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td></tr></table></td><td>Accept for first pair
<table><tr><th>number</th><th>flag</th></tr><tr><td>9</td><td>FALSE</td></tr><tr><td>9</td><td>FALSE</td></tr><tr><td>9</td><td>TRUE</td></tr></table>Do not award either mark point if first row is 9 FALSE 2</td><td>2</td></tr>
</table>

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Type MS · Write an algorithm on paper: flowcharts and pseudocode

Answer 1 of 6 · Question 6 part a · 6 marks

2025 JUN · Paper 01 · extract 3 of 4 · find part a in this mark scheme extract

Question	Answer	Additional Guidance	Mark
6(b)(iii)	Award **one** mark for: • To determine if a number is a prime number/none of the test numbers divide exactly into it (1)	Ignore reference to display the number	1
Question	**Answer**	**Additional Guidance**	**Mark**
6(b)(iv)	Award up to **two** marks for a linked explanation such as: • It makes the algorithm more efficient (1) because the loop will terminate as soon as a test number divides exactly into the number entered (1) • A count-controlled loop could make more comparisons/be less efficient (1) because it would not terminate as soon as the test number divides exactly into the number entered (1) • A count-controlled loop executes a fixed number of times/even after condition is met (1) post-condition loop stops when condition is met (1)	Explanation can be a combination of different marking points. E.g. MP1a and MP2b	2
		Total for question 6	**13**

Type MS · Write an algorithm on paper: flowcharts and pseudocode

Answer 1 of 6 · Question 6 part a · 6 marks

2025 JUN · Paper 01 · extract 4 of 4 · find part a in this mark scheme extract

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Type MS · Write an algorithm on paper: flowcharts and pseudocode

Answer 2 of 6 · Question 5 part b · 4 marks

2019 JUN · Paper 01 · extract 1 of 4 · find part b in this mark scheme extract

5(a)(i)	One mark for each correct row **Width** / **Length** / **Height** / **Output** 4 / 4 / 2 / Requires large container (1) 2 / 2 / 2 / Too small for cargo container (1) 3 / 8 / 5 / Too large for cargo container (1) Do not penalise spelling	3

Question Number	Answer	Additional Guidance	Mark
5(a)(ii)	Any response that indicates the concept of: Determining which size shipping container is needed for the indicated cargo.		1

Type MS · Write an algorithm on paper: flowcharts and pseudocode

Answer 2 of 6 · Question 5 part b · 4 marks

2019 JUN · Paper 01 · extract 2 of 4 · find part b in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
5(b)	SEND states [numStates] TO DISPLAY (1) or SEND state TO DISPLAY (1) SET numStates TO numStates + 1 (1) or LENGTH(states) (1) Concatenation of message and variable uses " and & & numStates & TO DISPLAY (1) or & LENGTH(states) & TO DISPLAY (1) or STRING(numStates) TO DISPLAY (1) Pseudocode is correct (1)	• Accept , or + in place of & • Ignore capitalisation of commands •	4

```
SET numStates TO 0
SET states TO ["France", "Singapore",
               "Malta", "Panama",
               "Greece", "Italy"]
FOR EACH state FROM states DO
    SEND state TO DISPLAY
    SET numStates TO numStates + 1
END FOREACH
SEND "There are " & numStates & "states" TO DISPLAY
```

```
SET numStates TO 0
SET states TO ["France", "Singapore",
               "Malta", "Panama",
               "Greece", "Italy"]
FOR EACH state FROM states DO
    SEND state TO DISPLAY
END FOREACH
SEND "There are " & LENGTH (states) & "states" TO DISPLAY
```

```
SET numStates TO 0
SET states TO ["France", "Singapore",
               "Malta", "Panama",
               "Greece", "Italy"]
FOR EACH state FROM states DO
    SEND state TO DISPLAY
    SET numStates TO numStates + 1
END FOREACH
SEND "There are " & STRING (numStates) TO DISPLAY
```

Type MS · Write an algorithm on paper: flowcharts and pseudocode

Answer 2 of 6 · Question 5 part b · 4 marks

2019 JUN · Paper 01 · extract 3 of 4 · find part b in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
5(c)(i)	Showing that index goes up to 4 and does not exceed 4 (1) Showing correspondence between values of index and loadWeight[index] (1)	• There does not have to be an explicit indication of the out-of-bounds / crashing error. • Ignore target and found columns • Allow two rows per loop with matching index and loadWeight values	2

target	found	index	loadWeight[index]
50500	FALSE	0	20000
		1	28000
		2	40000
		3	50000
		4	

target	found	index	loadWeight[index]
50500	FALSE	0	
		1	20000
		2	28000
		3	40000
		4	50000

Question Number	Answer	Additional Guidance	Mark
5(c)(ii)	1. Use of AND operator (in the test of relevant conditions) (1) 2. Use of suitable function to find the length of the array using a correct comparison to index (1) Example: WHILE ((NOT found) AND (index < LENGTH (loadWeight))) DO	• Do not penalise pseudocode syntax • Ignore changes to first condition as long as meaning is not lost	2

Type MS · Write an algorithm on paper: flowcharts and pseudocode

Answer 2 of 6 · Question 5 part b · 4 marks

2019 JUN · Paper 01 · extract 4 of 4 · find part b in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · Write an algorithm on paper: flowcharts and pseudocode

Answer 3 of 6 · Question 6 part a · 6 marks

2020 NOV · Paper 01 · extract 1 of 8 · find part a in this mark scheme extract

6(a)	Indicative content provided Other solutions do exist and should be awarded		6

Aspect of Solution	Marks			
	0	1	2	3
Functionality	No awardable content	There are significant errors in logic, leading to an overall solution that is non-functional	There are minor errors in logic, leading to an overall solution that is not completely functional	There are no errors in logic, leading to an overall solution that is fully functional
Accuracy of Notation	No awardable content	Notation follows a broadly unrecognisable convention that is applied inconsistently, although aspects of it are discernible	Notation follows a recognisable convention which is broadly discernible but is applied inconsistently	Notation follows a recognisable convention and is applied consistently throughout

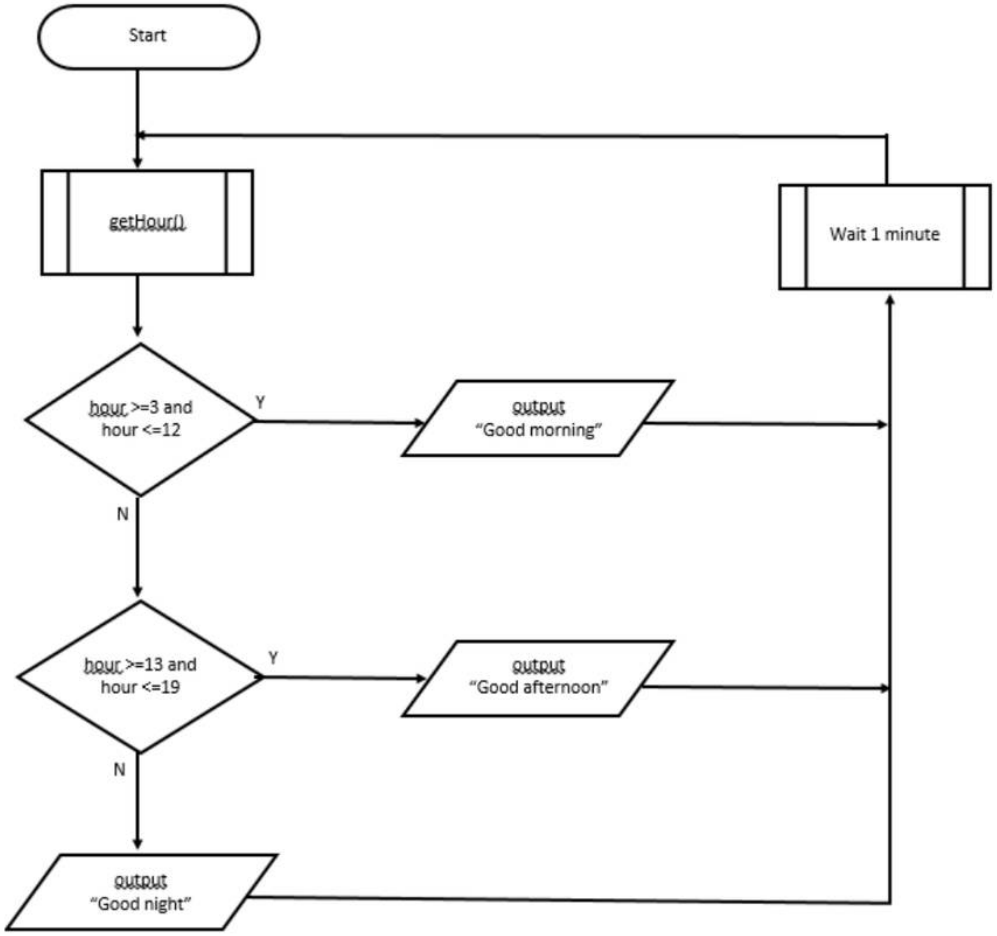
There are a maximum of 3 marks for functionality.
There are a maximum of 3 marks for accuracy of notation.
The marks for functionality and accuracy are awarded independently.

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Type MS · Write an algorithm on paper: flowcharts and pseudocode

Answer 3 of 6 · Question 6 part a · 6 marks

2020 NOV · Paper 01 · extract 2 of 8 · find part a in this mark scheme extract



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Type MS · Write an algorithm on paper: flowcharts and pseudocode

Answer 3 of 6 · Question 6 part a · 6 marks

2020 NOV · Paper 01 · extract 3 of 8 · find part a in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
6(b)(i)	• papaya needs changing (1) • lychee needs ordering (1)		2

Question Number	Answer	Additional Guidance	Mark
6(b)(ii)	7 (1)		1

Question Number	Answer	Additional Guidance	Mark
6(b)(iii)	Replace 8 with LENGTH(flavours)/LENGTH(volume) (1) or A new line 7: WHILE i < LENGTH(flavours) DO (1) or WHILE i < LENGTH(volume) DO (1)	Ignore case If new line is given award mark if logic is clear	1

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Type MS · Write an algorithm on paper: flowcharts and pseudocode

Answer 3 of 6 · Question 6 part a · 6 marks

2020 NOV · Paper 01 · extract 4 of 8 · find part a in this mark scheme extract

Question Number	Answer	Mark

20

Type MS · Write an algorithm on paper: flowcharts and pseudocode

Answer 3 of 6 · Question 6 part a · 6 marks

2020 NOV · Paper 01 · extract 5 of 8 · find part a in this mark scheme extract

6(c)	Indicative content: **Compiler** • One line of a compiled language maps to multiple executable instructions • Reads in a whole file and translates it at once • Produces an executable file • Executable file is portable between machines with the same architecture and operating systems • End-user cannot see the programming source code • Does not need an additional environment/software to run the code **Interpreter** • One line of a compiled language maps to multiple executable instructions • Reads, translates, and executes one line at a time • A special environment is needed to be installed on the user's machine to run the code; this is machine specific • Source code is portable to any machine with an interpreter which can run on it • End-user can see the programming source code **Similarities** • One line of a compiled language maps to multiple executable instructions/one to many relationship to machine code • Both are used to interpret high level programming languages • Both produce machine code from human-readable programming code **Differences** • Compiler reads in a whole file and translates it at once. Interpreter reads, translates and executes one line at a time • Compiler produces an executable file, interpreter does not • Executable files produced by a compiler will only execute on machines with the same architecture and operating systems. Source code produced by an interpreter is portable to any machine with an interpreter which can run on it • Compiled code does not require any additional software	6

Type MS · Write an algorithm on paper: flowcharts and pseudocode

Answer 3 of 6 · Question 6 part a · 6 marks

2020 NOV · Paper 01 · extract 6 of 8 · find part a in this mark scheme extract

	to execute the code. An interpreter needs a special environment to be installed on the user's machine to run the code; this is machine specific	

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Type MS · Write an algorithm on paper: flowcharts and pseudocode

Answer 3 of 6 · Question 6 part a · 6 marks

2020 NOV · Paper 01 · extract 7 of 8 · find part a in this mark scheme extract

Level	Mark	Descriptor
	0	No rewardable content.
Level 1	1–2	Basic, independent points are made showing elements of knowledge and understanding of key concepts/principles of computer science. The discussion will contain basic information with little linkage between points made.
Level 2	3-4	Demonstrates adequate knowledge and understanding of key concepts/principles of computer science. The discussion shows some linkages and lines of reasoning with some structure.
Level 3	5-6	Demonstrates comprehensive knowledge and understanding by selecting relevant knowledge and understanding of key concepts/principles of computer science to support the discussion being presented. The discussion shows a well-developed, sustained line of reasoning which is clear, coherent, and logically structured.

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Type MS · Write an algorithm on paper: flowcharts and pseudocode

Answer 3 of 6 · Question 6 part a · 6 marks

2020 NOV · Paper 01 · extract 8 of 8 · find part a in this mark scheme extract

Type MS · Write an algorithm on paper: flowcharts and pseudocode

Answer 4 of 6 · Question 4 part c · 5 marks

2021 NOV · Paper 01 · extract 1 of 3 · find part c in this mark scheme extract

<table>
<tr>
<td>4(a)</td>
<td>Award one mark from:

• a step-by-step description of a process that completes a task (1)
• a set of instructions that describes how to get something done (1)</td>
<td></td>
<td>1</td>
</tr>
<tr>
<td>4(b)(i)</td>
<td>Award one mark each to a maximum of five marks for:

• two correct decisions in last two lines (1)
• two correct outputs (1)
• correct increment at Count=Count+1 (1)
• correct change for Number3=Number2+Number1 (1)
• correct change for Number1=Number2 AND Number2=Number3 (1)

<table>
<tr><th>Number 1</th><th>Number 2</th><th>Number 3</th><th>Count</th><th>Output</th><th>Count = 2?</th></tr>
<tr><td>1</td><td>1</td><td>-</td><td>0</td><td>1 1</td><td>False</td></tr>
<tr><td>1</td><td>2</td><td>2</td><td>1</td><td>2</td><td>False</td></tr>
<tr><td>2</td><td>3</td><td>3</td><td>2</td><td>3</td><td>True</td></tr>
</table>
</td>
<td>Ignore spelling errors

Allow true false, Y N etc. for decisions</td>
<td>5</td>
</tr>
</table>

Type MS · Write an algorithm on paper: flowcharts and pseudocode

Answer 4 of 6 · Question 4 part c · 5 marks

2021 NOV · Paper 01 · extract 2 of 3 · find part c in this mark scheme extract

4(b)(ii)	Award **one** mark each for: • helps visualise how the algorithm works (1) • helps detect (logic) errors (1)		1

Question Number	Answer	Additional Guidance	Mark
4(c)	Award **one** mark each to a maximum of **five** marks for: • SET count TO 0 (1) • IF statement checks for a match to 1 vowel (1) • IF statement checks for a match to all vowels (1) • count incremented correctly (1) • display count (1) • concatenates suitable message with the count (1) SET word TO "elephant" SET count TO 0 FOR EACH letter FROM word DO IF letter = 'a' OR letter = 'e' OR letter = 'i' OR letter = 'o' OR letter = 'u' THEN SET count TO count + 1 END IF END FOREACH SEND 'The number of vowels is ' & count TO DISPLAY		5
		Total	**12**

Type MS · Write an algorithm on paper: flowcharts and pseudocode

Answer 4 of 6 · Question 4 part c · 5 marks

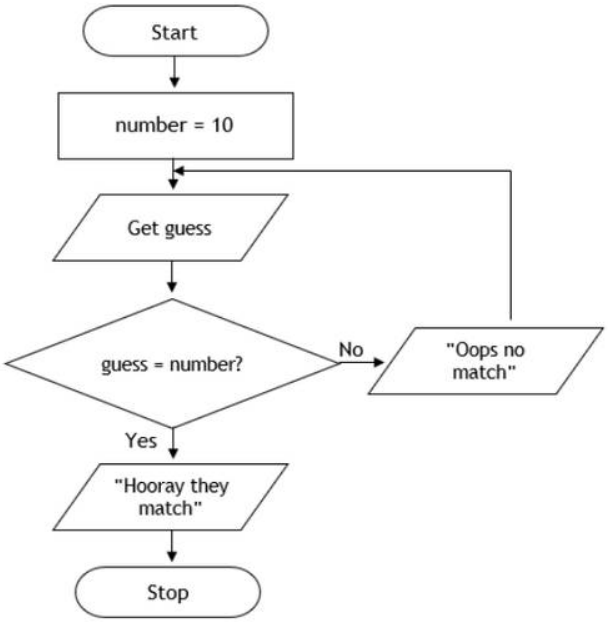
2021 NOV · Paper 01 · extract 3 of 3 · find part c in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · Write an algorithm on paper: flowcharts and pseudocode

Answer 5 of 6 · Question 4 part a.i · 5 marks

2022 JUN · Paper 01 · extract 1 of 3 · find part a.i in this mark scheme extract

4(a)(i)	Award up to **five** marks for: • Start and stop terminators in the correct positions (1) 1-3 • Number set to 10 AND Get guess in the correct positions (1) 4-6 • Loop back to before Get guess and after number set to 10 if there is no match (1)Does not need to go via message box • Yes/no labels on decision match output messages (1) 4-6 • Correctly connected as in MS image, with at least 6 arrows correct (1) 7-9  	Boxes should be marked by content rather than shape.	5
4(a)(ii)	**The only correct answer is C** *A is not correct because a simulation is a completed program* *B is not correct because a cipher is a form of encryption* *D is not correct because a truth table is a method of testing an algorithm*		1

Type MS · Write an algorithm on paper: flowcharts and pseudocode

Answer 5 of 6 · Question 4 part a.i · 5 marks

2022 JUN · Paper 01 · extract 2 of 3 · find part a.i in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
4(b)(i)	Award **one** mark for: • D		1
4(b)(ii)	Award **one** mark for: • B		1

Question Number	Answer	Additional Guidance	Mark
4(c)(i)	Award up to **four** marks for: • RedPoints column correct (1) • OrangePoints AND NumOranges columns correct (1) • Score correct in row 8 OR row 9 (1) -1 or below • Outputs correct and starting on the same row as the score OR the row below (1) can all be on one line		4
4(c)(ii)	Award **one** mark for: • 23		1

Colour	Score	Red Points	Orange Points	Num Oranges	Outputs
	0	0	0	0	
red		1			
orange			8	1	
red		2			
red		3			
orange			16	2	
-1					
	19				
					Score: 19
					Number of reds: 3
					Number of oranges: 16

Type MS · Write an algorithm on paper: flowcharts and pseudocode

Answer 5 of 6 · Question 4 part a.i · 5 marks

2022 JUN · Paper 01 · extract 3 of 3 · find part a.i in this mark scheme extract

4(c)(iii)	Award **one** mark for: • Pseudododode that replaces OrangePoints with NumOranges on line 23 (1) SEND ("Number of oranges: "& NumOranges) TO DISPLAY		1
		Total for question 4	**14**

Question Number	Answer	Additional Guidance	Mark

Type MS · Write an algorithm on paper: flowcharts and pseudocode

Answer 6 of 6 · Question 6 part d · 6 marks

2024 JUN · Paper 01 · extract 1 of 5 · find part d in this mark scheme extract

6(a)	**The only correct answer is D** *A is not correct because it is not a REPEAT UNTIL / DO WHILE loop* *B is not correct because it is not a FOR loop* *C is not correct because there is no such loop*		1

6(b)(i)	Award **one** mark for any of the following:
	
	• `WHILE NOT (index >= LENGTH(arrival))`
	• `WHILE NOT (index > LENGTH(arrival)-1)`
	• `WHILE index < LENGTH(arrival)`
	• `WHILE index <= LENGTH(arrival)-1`
		Allow hard coded values for the number of items in the list.
		For example,
		`index >= 7`
		`index > 6`
			1

6(b)(ii)	Award **one** mark for any of the following: • `IF arrival[index] > 0` • `IF arrival[index] >= 1`		1
6(b)(iii)	Award **one** mark for: • `late` is incremented (1) `SET late TO late + 1`		1

Type MS · Write an algorithm on paper: flowcharts and pseudocode

Answer 6 of 6 · Question 6 part d · 6 marks

2024 JUN · Paper 01 · extract 2 of 5 · find part d in this mark scheme extract

6(b)(iv)	Award **one** mark for:
	
	• On time is calculated as the remainder when early and late trains are subtracted from the total number of trains (1)
	`LENGTH(arrival) - (late + early)`
		Allow number of trains to be hard coded as 7 in place of
		`LENGTH(arrival)`
		
		Allow equivalent expressions:
		`LENGTH(arrival) - late - early`
			1

6(c)	Award **one** mark for any of the following: • Otherwise there would be an infinite loop (1) • It is used to index all elements in the (`arrival`) array (1) • To access the next value in the (`arrival`) array on the next iteration (1) • So that the condition used in the while loop evaluates to True when all the elements in arrival have been processed (1)		1

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Type MS · Write an algorithm on paper: flowcharts and pseudocode

Answer 6 of 6 · Question 6 part d · 6 marks

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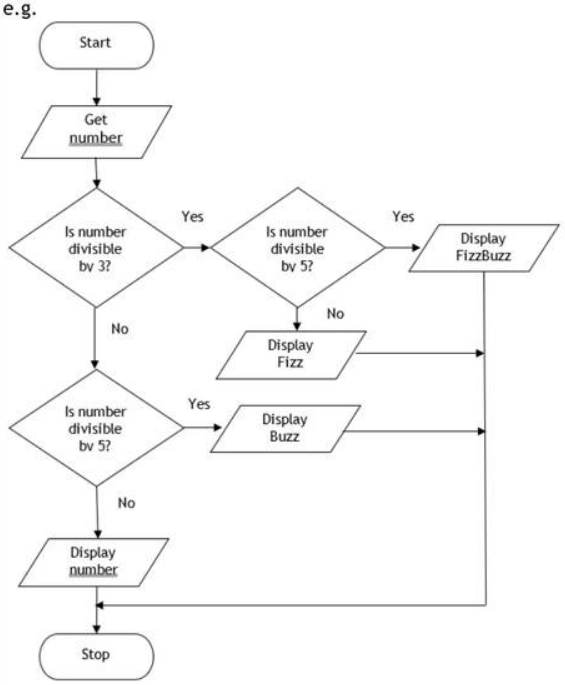
6(d)	Award **one** mark for each up to a maximum **six** marks: • Get number immediately follows start (1) • Number divisible by 3 and number divisible by 5 → Display Fizz (1) • Number divisible by 3 and number divisible by 5 → Display Buzz (1) • Number divisible by 3 and number divisible by 5 → Display FizzBuzz (1) • Number not divisible by 3 and number not divisible by 5 → Display n (1) • The flowchart represents a fully functional solution (1)	For last mark point: All symbols fully connected with directed arrows. Start and stop terminators used correctly. No hanging symbols. Decision boxes have exactly two labelled outputs For each of the first five mark points the mark should only be awarded if there is one correct logical path to get to the outcome. DO NOT credit a point if additional symbols have been added.	

Type MS · Write an algorithm on paper: flowcharts and pseudocode

Answer 6 of 6 · Question 6 part d · 6 marks

2024 JUN · Paper 01 · extract 4 of 5 · find part d in this mark scheme extract

e.g.



			6
		Total for question 6	**12**

Type MS · Write an algorithm on paper: flowcharts and pseudocode

Answer 6 of 6 · Question 6 part d · 6 marks

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